“Oh no, I lost them somewhere else, but the light is much better over here.” In the quantum-gravity game, AdS/CFT is the world’s brightest lamppost. By studying it we’ve uncovered a large number of fascinating concepts that are useful to theoretical physicists, but there is no direct route to using that knowledge to understand why apples fall from trees, or other aspects of gravity in the space around us. It’s worth continuing the pursuit, but important to keep our eyes on the prize: understanding the world in which we actually live.

The implications of holography for real-world black holes are less clear than they are for the imaginary world of AdS/CFT. Are we saying that classical general relativity was completely wrong about the interior of a black hole appearing empty, and that in fact an infalling observer would smack into a holographic surface upon encountering the event horizon? We are not—at least, most adherents of holography aren’t saying that. Rather, they appeal to a related and equally startling idea, *black-hole complementarity*. It was proposed by Susskind and others, using terminology that intentionally recalls Bohr’s philosophy of quantum measurement.

The black-hole version of complementarity says that things are a little more nuanced than simply “the interior of a black hole looks like ordinary empty space” or “all the information about the black hole is encoded on the event horizon.” In fact both are true, but we can’t speak both languages at the same time. Or, as physicists are more likely to put it, they don’t simultaneously appear true to any single observer. To an observer falling through the event horizon, everything looks like normal empty space, while to an observer looking at the hole from far away, all of the information is spread across the horizon.

Even though this behavior is fundamentally quantum-mechanical, it does have a classical precursor. Think about what happens to a book (or a star, or whatever) when we throw it into a black hole in classical general relativity. From the book’s point of view, it just passes right into the interior. But the effect of spacetime warping is strong near the event horizon, so that’s not what an external observer would see. They would see the book